Vacuum Rabi oscillation in coupled highly-dissipative cavity quantum electrodynamics

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Abstract: We demonstrate strong anharmonicity of the polariton dressed states in a highly dissipative cavity quantum electrodynamics system via dark state resonances. Vacuum Rabi oscillation and photon blockade occur even for decay-to-interaction rate ratio exceeding 100.
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Cavity quantum electrodynamics (QED) [1] provides a critical resource for quantum information processing. To coherently manipulate single and coupled qubits, a key prerequisite is to reach the strong coupling regime, where the emitter-field coupling strength exceeds the decay rates of the dipole emitter and the photon field in the cavity. In the past two decades great efforts have been made to improve the quality (Q) factor and reduce the mode volume (V) of the optical resonators for stronger interactions, using Fabry-Pérot cavities [2], Bragg cavities [3], whispering-gallery mode cavities [4], photonic crystal cavities [5], and hybrid plasmonic-photonic cavities [6, 7]. However, it remains difficult to achieve high quality factor and small mode volume simultaneously. For example, whispering-gallery mode cavities possess ultrahigh Q factors, while the mode volumes are relatively large. For photonic crystal cavities, subwavelength light confinement can be realized whereas the Q factors are relatively low.

Unlike efforts to improve the $Q/\sqrt{V}$ figures-of-merit of an optical cavity, here we propose to reach the strong coupling regime via dark state resonances, which removes the requirement for high $Q/\sqrt{V}$ factors of the cavities. We take advantage of the coupled cavity system to achieve strong dark state interaction between a single quantum emitter and a high-Q but large V optical mode, mediated optically by a low-Q optical mode. We demonstrate that vacuum Rabi oscillations and anharmonicity in the polariton dressed states occur even for bosonic decay-to-interaction rate ratio exceeding 100. It offers great potential for coherent operations of single photons and polaritons.

As shown in Fig. 1(a), a cavity QED system, consisting of a dipole quantum emitter and an optical cavity, is coupled to an auxiliary cavity through a single-mode optical waveguide. Here κ_1, κ_2 and γ represent the decay rates of mode a_1, a_2 and the emitter; g denotes the emitter-field coupling strength between the emitter and mode a_1; J describes the inter-cavity coupling strength between mode a_1 and a_2 through the single-mode optical waveguide. We show that the highly dissipative cavity QED systems ($\kappa_1 >> g$) can be turned into the strong coupling regime with the effective coupling between mode a_2 and the emitter, as illustrated by the energy level diagrams in Figs. 1(b) and (c).

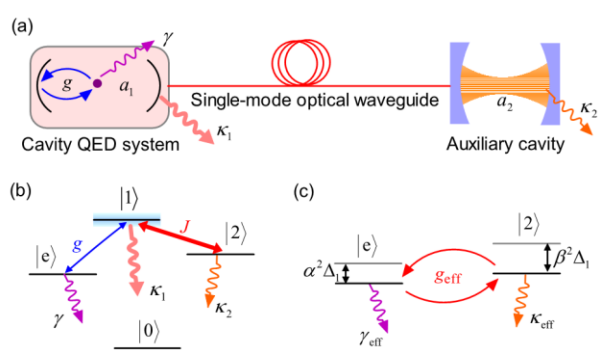

Fig. 1. (a) Schematic of the cavity QED system coupled to an auxiliary optical cavity. (b) Energy level diagram of the coupled system. (c) Dark state interaction after eliminating state $|1\rangle$.

In the time domain, vacuum Rabi oscillation is a direct evidence of the coherent energy exchange between the emitter and the cavity photon field. Here we numerically solve the quantum master equation to obtain the exact results. We assume initially the emitter is in the excited state and the two optical modes are in their vacuum states, then we obtain the exact numerical results for the time evolution of the mean photon numbers $N_1(t)$, $N_2(t)$ and the probability for the emitter being in the excited state $P_e(t)$. As shown in Figs. 2(a) and 2(b), even for large κ_1/g, vacuum Rabi oscillation phenomenon occurs for several periods, revealing that the decoherence time is much longer than the energy exchange period. This is in contrary to the case without the auxiliary cavity (insets of Figs. 2(a) and

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2(b)), where the emitter exponentially decays from the excited state. Note that mode a_1 keeps at the vacuum state (green triangles), while the photon number in mode a_2 oscillates (blue open circles). This reveals that the interaction is between the emitter and mode a_2, while mode a_1 only act as a background. The analytical results for the emitter's occupancy in the excited state (red solid curve) are in good accordance with the exact numerical results (red closed circles).

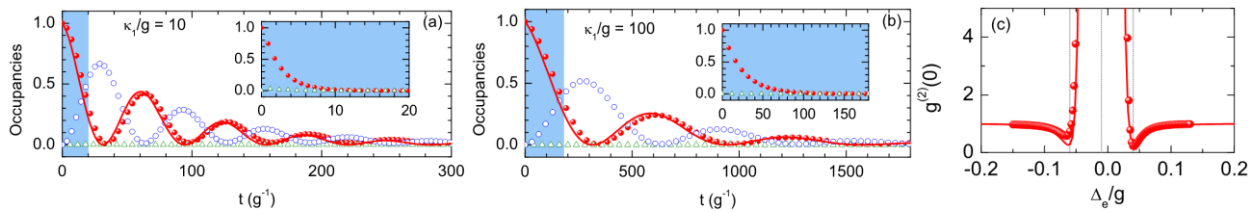

Fig. 2. Time evolution of the mean photon numbers $N_1(t)$ (green triangles), $N_2(t)$ (blue open circles) and the probability for the dipole emitter being in the excited state $P_e(t)$ (red closed circle) for $\kappa_1/g = 10$ (a) and $\kappa_1/g = 100$ (b). The red solid curve corresponds to the analytical results. Insets: Comparative $N_1(t)$ (green triangles) and $P_e(t)$ (red closed circle) without the auxiliary cavity ($J = 0$). (c) Second order correlation function $g^{(2)}(0)$ as a function of the probe-emitter detuning Δ_e.

The effective strong coupling offers great potential for single-photon manipulation and quantum logic gate operation. For example, photon blockade phenomenon occurs in this coupled system. The anharmonicity of the polariton dressed states level spacing, from the first and second Jaynes-Cummings rungs, results in the suppression of two-photon transmission. Therefore, due to the strong anharmonicity, photon blockade of the second photon by the first photon can occur. This is quantitatively characterized by the zero-delay second-order correlation function $g^{(2)}(0)$. We use a weak probe laser to obtain the exact results of $g^{(2)}(0)$ numerically. In Fig. 2(c) $g^{(2)}(0)$ as a function of the probe-emitter detuning Δ_e is plotted. It reveals that $g^{(2)}(0)$ approaches 0 for certain detunings Δ_e, indicating strong antibunching effect and sub-Poissonian photon statistics. The analytical results obtained from the effective parameters agree well with the exact results. Therefore, to realize the photon blockade, we only need a cavity coupled to a bad-cavity QED system, while good-cavity QED system is not necessary.

It should be noted that, although the auxiliary cavity is required to be high-Q ($\kappa_2 < g$), it does not need to interact directly with the quantum emitter, and the mode volume is not necessary to be small. Therefore, the scheme does not require a high figure of merit $Q/\sqrt{V}$ for the auxiliary cavity. Together with the allowed low Q factor for the primary cavity, both the two cavities can be low in figure of merit $Q/\sqrt{V}$.

In summary, we have presented a protocol for realizing effective coherent polariton states in a highly-dissipative cavity QED system. By employing the coupled cavity configuration, we show that a highly dissipative cavity interacting simultaneously with a single quantum emitter and an auxiliary optical cavity leads to the dark state resonance between the quantum emitter and the auxiliary cavity. It is demonstrated that effective strong coupling can be achieved even with low $Q/\sqrt{V}$ cavities, with prominent vacuum Rabi oscillation and ladder anharmonicity phenomena for photon blockade. The cavity coupling to the quantum emitter can be highly dissipative even with bosonic decay rate κ_1/g in excess of 100. The system enables single photon manipulation like photon blockade and quantum logic gate operations. This approach offers opportunities to exploit both theoretical and experimental physics in the strong light-matter interaction regime without stringent cavity requirements.

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